

9 The chloroplasts of leaves of tobacco plants, *Nicotiana* sp., contain chlorophyll *a* and chlorophyll *b*.

(a) Describe the role of chlorophyll *b* in photosynthesis.

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(b) A mutant tobacco plant was found to contain more chlorophyll *b* than normal tobacco plants.

An investigation was carried out to measure the rate of photosynthesis of normal and mutant tobacco plants at increasing light intensities. The rate of production of oxygen was used as a measure of the rate of photosynthesis.

Other variables were kept constant.

Fig. 9.1 shows the results of this investigation.

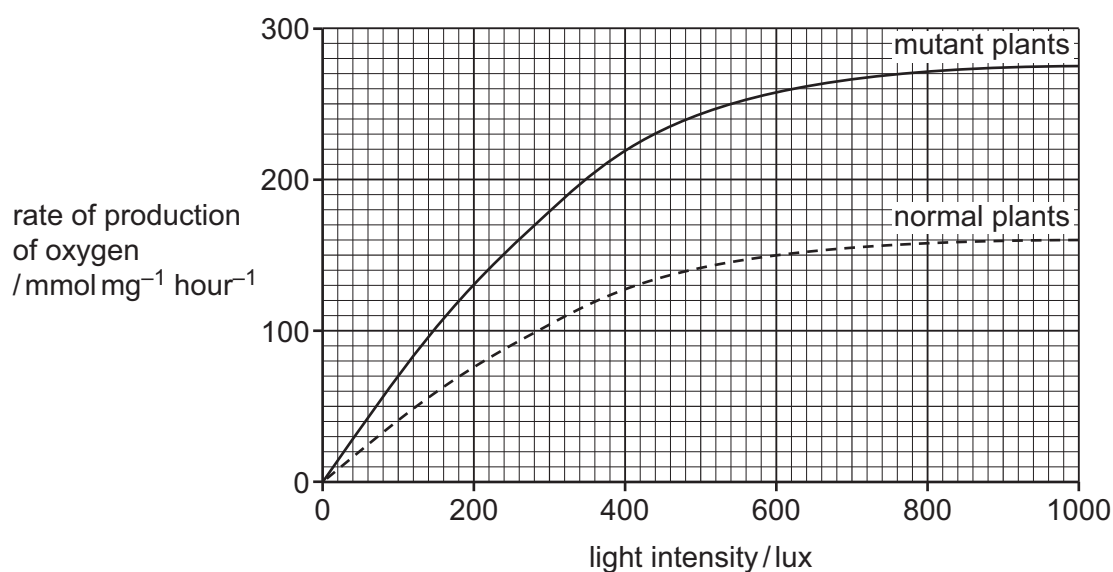


Fig. 9.1



(i) Describe the results shown in Fig. 9.1.

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(ii) It was observed that the mutant tobacco plants had a faster growth rate than the normal tobacco plants.

Suggest explanations for this observation.

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[Total: 9]